MIL-T-52532B(ME)
5 August 1981
SUPERSEDING
MIL-T-52532A(ME)
21 November 1975

MILITARY SPECIFICATION

TRAINING AID, PROTRACTORS, CIRCULAR (LARGE AND SMALL), MAP READING

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of training aid circular protractors.

1.2 Classification. Training aid circular protractors covered by this specification shall be the following types, as specified (see 6.2):

Type I - Large - 5 feet diameter.
Type II - Small - 18 inch diameter.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

* U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/6106

FSC 6910

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self- addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:24Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

D13211E7408	- Training Aid, Protractor, Circular, Large.
D13211E7409	- Training Aid, Protractor, Circular, Small.
B13211E7423	- Latch.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The protractors shall be as shown on the applicable drawings and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:24Z
Check the source to verify that this is the current version before use.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the two types of training aid circular protractors. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.3 Type I, large. Type I protractors shall be as shown on Drawing D13211E7408 and as specified herein.

3.3.1 Latch. The latch for Type I protractors shall be as shown on Drawing B13211E7423 and as specified herein.

3.4 Type II, small. Type II protractors shall be as shown on Drawing D13211E7409 and as specified herein.

3.5 Identification marking. The protractors shall be identified in accordance with MIL-STD-130.

3.6 Workmanship. Edges and adjacent surfaces shall be free from loose or peeled fibers and splinters. Metal components shall be free from sharp edges and burrs. All screws shall be fully turned down.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:24Z
Check the source to verify that this is the currentversion before use.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The protractors shall be examined herein for the following major defects:

101. Dimensions not as specified.
102. Components not assembled as specified, or missing.
103. Materials not as specified.
104. Treatment and painting not as specified.
105. Identification marking not as specified.
106. Workmanship not as specified.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

107. Materials, methods, and containers not as specified. Each incorrect material, method, or container shall be considered one defect.
108. Protractors bent.
109. Protractors not cushioned or painted surfaces not protected as specified for Level A.
110. Boxes not closed and sealed as specified for Level A.
111. Strapping not as specified for Level A.
112. Boxes not reinforced with tape as specified for Level B.
113. Marking missing, illegible, incorrect, or incomplete.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:24Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each Type I protractor shall be folded. Bending of the protractor shall not be permitted. The protractor shall be secured and cushioned to prevent damage to the graduated painted surfaces. The cushioned protractor shall be placed in a box conforming to PPP-B-636, W5c, style optional. The box shall be closed and sealed as specified for Method V in accordance with the appendix to the box specification. Each Type II protractor shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3. Bending of the protractor shall not be permitted. The protractor shall be cushioned to prevent damage to the bag and the graduated painted surface.

5.1.1.1 Consolidation. The Type II protractors shall be consolidated in a close-fitting box conforming to PPP-B-636, W5c, style optional. The gross weight shall not exceed the limitation of the box specification.

5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188. Bending of protractors shall not be permitted.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Protractors of like description, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional. The box shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable; and, unless otherwise specified herein, strapping shall be Finish B. When specified (see 6.2), Finish A strapping shall be used.

5.2.2 Level B. Protractors of like description, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-640, Class 2, style optional, or PPP-B-636, V3c, style optional. The size and weight limitations shall conform to the box specification. The box shall be reinforced with bands of reinforced filament tape applied in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.3 Marking. Marking for military levels of protection shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The protractors are components of a map reading set and are used as aids in teaching basic map reading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:24Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of protractor required (see 1.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When strapping shall be Finish A (see 5.2.1).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

Custodian:
Army - ME

Review activity:
DOD - NS

Preparing activity:
Army - ME

Project. 6910-A407

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:24Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
USA Mobility Equipment R&D Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-T-52532B(ME) Training Aid, Protractors, Circular (Large and Small), Map Reading

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:24Z
Check the source to verify that this is the current version before use.